

① $\lim_{n \rightarrow \infty} \frac{u_n}{v_n}$ (Comparing test) $p > 1$ Convergent
 is non zero / integer if it is of the form $\frac{1}{n^p}$ $p \leq 1$ divergent

② in G.P. $r < 1 \rightarrow$ convergent
 $r \leq -1 \rightarrow$ oscilary
 $r > 1 \rightarrow$ divergent

③ D' Alembert's Ratio Test :-
 $\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = k$

$k < 1 \rightarrow$ convergent

$k > 1 \rightarrow$ Divergent

$k = 1 \rightarrow$ Test fail

④ $\sum_{n=1}^{\infty} n^3 \Rightarrow S_n = \frac{n^2(n+1)^2}{4}$

⑤ Raabe Test

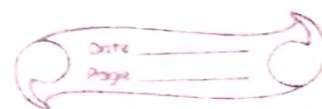
$$\lim_{n \rightarrow \infty} n \left[\frac{u_n}{u_{n+1}} - 1 \right] = k$$

$k > 1 \rightarrow$ Convergent

$k < 1 \rightarrow$ Divergent

$k = 1 \rightarrow$ Test fail

β & γ function



$$(1) \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx \quad (n > 0)$$

$$(2) \Gamma(n+1) = \int_0^{\infty} e^{-x} x^n dx$$

$$(3) \Gamma(n+1) = n \Gamma(n) \quad \& \quad \Gamma(n) = (n-1) \Gamma(n-1)$$

$$(4) \Gamma(n+1) = n!$$

$$(5) \Gamma(n) \quad ; \quad n \in \text{Positive Integer} \quad ; \quad (n-1)!$$

$$\Gamma(n) \quad ; \quad n \in \text{negative Integer} \quad ; \quad \infty$$

$$n \in 0 \quad ; \quad \infty$$

$$n \in 1 \quad ; \quad 1$$

$$\Gamma(n) \quad ; \quad n \in \text{fraction} \quad ; \quad \Gamma\left(\frac{3}{2}\right) = \left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)$$

$$\Gamma(n) \quad ; \quad n \in \text{Negative fraction} \quad ; \quad \Gamma(n) = \frac{\Gamma(n+1)}{n}$$

$$\text{eg: } \Gamma\left(-\frac{5}{3}\right) = \frac{\Gamma\left(-\frac{2}{3}\right)}{\left(-\frac{5}{3}\right)} = \frac{\Gamma\left(\frac{1}{3}\right)}{\left(-\frac{5}{3}\right)\left(-\frac{2}{3}\right)} = \frac{\Gamma\left(\frac{1}{3}\right)}{\frac{10}{9}}$$

$$(6) \text{ Second form of } \Gamma \text{ function: } \Gamma(n) = 2 \int_0^{\infty} e^{-x^2} x^{2n-1} dx$$

$$(7) \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(\frac{1}{4}\right) \cdot \Gamma\left(\frac{3}{4}\right) = \sqrt{2\pi}$$

$$\Gamma\left(\frac{3}{4}\right) \cdot \Gamma\left(\frac{5}{4}\right) = \frac{\pi}{2\sqrt{2}}$$

$$\Gamma\left(\frac{5}{4}\right) \cdot \Gamma\left(\frac{7}{4}\right) = \frac{\pi}{16} \sqrt{2\pi}$$

$$(8) \text{ Prove } \int_0^{\infty} \frac{x^q}{a^x} dx = \frac{1}{(\log a)^{q+1} \Gamma(q+1)}$$

$$\text{Put } a^x = e^t \Rightarrow x \log a = t$$

$$(9) e^{-i\theta} = \cos\theta - i \sin\theta$$

$$(10) (i) \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$(ii) \beta(m+1, n+1) = \int_0^1 x^m (1-x)^n dx$$

$$(iii) \beta(m, n) = \beta(n, m)$$

$$(iv) \beta(m, n) = \frac{\sqrt{m} \cdot \sqrt{n}}{\sqrt{m+n}} = \frac{(m-1)! (n-1)!}{(m+n-1)!}$$

m, n is integer

(v) Second form of β function:-

$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta$$

$$(vi) \int_0^{\infty} \beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx \quad \&$$

$$\beta(m+1, n-m-1) = \int_0^{\infty} \frac{x^m}{(1+x)^n} dx$$

⑩ Duplication formula:-

$$(i) \int_0^{\pi/2} \sin^p \theta \cdot \cos^q \theta d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

or

$$\int_0^{\pi/2} \sin^p \theta \cdot \cos^q \theta d\theta = \frac{1}{2} \frac{\left|\frac{p+1}{2}\right| \left|\frac{q+1}{2}\right|}{\left|\frac{p+q+2}{2}\right|}$$

PARTIAL DIFFERENTIATION

$$(1) \frac{\partial}{\partial x} [\sin f(x, y, z)] = \cos f(x, y, z) \cdot \frac{\partial f}{\partial x}$$

$$(2) \frac{\partial}{\partial x} [\tan f(x, y, z)] = \sec^2 f(x, y, z) \frac{\partial f}{\partial x}$$

$$(3) \frac{\partial}{\partial x} [e^{f(x, y, z)}] = e^{f(x, y, z)} \cdot \frac{\partial f}{\partial x}$$

$$(4) \frac{\partial}{\partial x} (u \pm v) = \frac{\partial u}{\partial x} \pm \frac{\partial v}{\partial x} \quad \begin{matrix} u f(x, y) \\ v f(x, y) \end{matrix} \quad \frac{\partial}{\partial x}$$

$$(5) \frac{\partial}{\partial x} (u \cdot v) = u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x}$$

$$(6) \frac{\partial}{\partial x} \left(\frac{u}{v} \right) = \frac{v \frac{\partial u}{\partial x} - u \frac{\partial v}{\partial x}}{v^2} \quad (v \neq 0)$$

$$(7) \frac{\partial}{\partial x} (kx) = k \frac{\partial}{\partial x} (x)$$

$$(8) \frac{\partial (k)}{\partial x} = 0$$

$$(9) \frac{\partial}{\partial y} \left(\frac{1}{f(x, y, z)} \right) = \frac{-1}{[f(x, y, z)]^2} \frac{\partial f}{\partial y}$$

$$(10) \frac{\partial}{\partial x} (\sqrt{f(x, y, z)}) = \frac{1}{2\sqrt{f(x, y, z)}} \frac{\partial f}{\partial x}$$

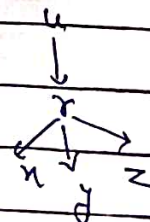
(11) Composite function:-

Type-I:- If $u = f(r)$ & $r = \phi(x, y, z)$

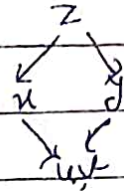
$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial z}$$



⑨ Type-II:- If $z = f(x, y)$, $x = g(u, v)$ $y = h(u, v)$



$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

⑩ Implicit function:- $\frac{\partial y}{\partial x} = - \frac{(\frac{\partial f}{\partial x})}{(\frac{\partial f}{\partial y})}$

⑪ $\frac{\partial (\tan^{-1} x)}{\partial x} = \frac{1}{1+x^2}$

⑫

Euler's theorem:-

if $F(x, y)$ be a homogeneous function in x and y of degree n , then,

(i) $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f$

(ii) $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f$

Fourier Series

$$(1) f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$(2) z = \cos \theta + i \sin \theta$$

$$(3) \int u \cdot v dx = u \int v dx - \int \left[\int v dx \left(\frac{du}{dx} \right) \right] dx + C$$

$$= uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

$$(4) \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$(5) \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

Types of limits

↓	↓	↓	↓
Type-(1)	Type-(2)	Type-(3)	Type-(4)
$(0-2\pi)$	$(-\pi \text{ to } \pi)$	$(0 \text{ to } 2l) /$ $(0 \text{ to } 2a)$	$(-l \text{ to } l) /$ $(-a \text{ to } a)$

Type-(1) & (2) के लिए:-

$$(1.1) a_0 = \frac{1}{2\pi} \int_c^{c+2\pi} f(x) dx$$

$$(1.2) a_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos nx dx$$

$$(1.3) b_n = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin nx dx$$

Type-(3) & (4) के लिए:-

$$(6) f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \left(\frac{n\pi x}{l} \right) + \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{l} \right)$$

where $l = \frac{\text{Final limit} - \text{initial limit}}{2}$

$$(6.1) \quad a_0 = \frac{1}{2l} \int_c^{c+2l} f(x) dx$$

$$(6.2) \quad a_n = \frac{1}{l} \int_c^{c+2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$(6.3) \quad b_n = \frac{1}{l} \int_c^{c+2l} f(x) \sin \frac{n\pi x}{l} dx$$

(7) Half Range cosine & sine series for limit $(-l \text{ to } l)$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$(7.1) \quad a_0 = \frac{1}{l} \int_0^l f(x) dx$$

$$(7.2) \quad a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

$$(7.3) \quad b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

where, $\boxed{l = F - I}^*$

(7.4) Half Range cosine series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$$

(7.5) Half Range sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

(8)

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

Indefinite Integration

PAGE NO.: 61

★★ Formula Set-1: Using Definition,

$$1) \int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$2) \int \frac{1}{x} dx = \log_e |x| + C$$

$$3) \int e^x dx = e^x + C$$

$$4) \int a^x dx = \frac{a^x}{\log_e a} + C$$

$$5) \int \sin x dx = -\cos x + C$$

$$6) \int \sec^2 x dx = \tan x + C$$

$$7) \int \cos x dx = \sin x + C$$

$$8) \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$9) \int \sec x \tan x dx = \sec x + C$$

$$10) \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$11) \int \tan x dx = -\log_e |\cos x| + C = \log_e |\sec x| + C$$

$$12) \int \cot x dx = \log |\sin x| + C$$

$$13) \int \operatorname{cosec} x dx = \log |\operatorname{cosec} x - \cot x| + C$$

$$14) \int \sec x dx = \log |\sec x + \tan x| + C = \log \left| \tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right| + C$$

$$15) \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C; |x| \leq 1$$

$$16) \int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} |x| + C; |x| > 1$$

17. $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$

** Theorems of Integration:-

1. $\int k \cdot f(x) dx = k \int f(x) dx$, where k is any constant.
2. $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$.
3. If $\int f(x) dx = F(x)$
then, $\int f(ax \pm b) dx = \frac{1}{a} F(ax \pm b)$.

** Questions involving Trigonometry:-

Q. $\int \left(\frac{\sin^6 x + \cos^6 x}{\tan^2 x} \right) \sec^2 x \cdot dx$

Ans $\Rightarrow \int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} \cdot dx \Rightarrow \int \frac{(\sin^2 x)^3 + (\cos^2 x)^3}{\sin^2 x \cdot \cos^2 x} dx$

$\Rightarrow \int \frac{1 - 3 \cdot \sin^2 x (\cos^2 x)(1)}{\sin^2 x \cos^2 x} dx \Rightarrow \int \frac{1 - 3 \sin^2 x \cos^2 x}{\sin^2 x \cos^2 x} dx$

$\Rightarrow \int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} - 3x + C \Rightarrow \int \sec^2 x \frac{dx}{dx} + \int \csc^2 x \frac{dx}{dx} - 3x + C$

$\Rightarrow \tan x - \cot x - 3x + C$

** Methods of Integration:-

1. Substitution
2. By parts
3. Partial fractions

*** Formula Set-2: - Based on Substitution:-

① 1. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$

⑤ 2. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

② 3. $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log |x + \sqrt{x^2 + a^2}| + C$

③ 4. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log |x + \sqrt{x^2 - a^2}| + C$

⑥ 5. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$